

Non-Dominant Per-Block Projection in CPSV16

§II Step 1

under the One-Copy Normal-Canonical BNT Formulation

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1 Notation

Fix two families of MPS site tensors $\{A_j\}_{j=1}^{r_A}$ and $\{B_k\}_{k=1}^{r_B}$ with nonzero scalars $\mu_j^A, \mu_k^B \in \mathbb{C}$. The assembled MPV at length N is

$$V^{(N)}(A_{\text{tot}}) = \sum_{j=1}^{r_A} (\mu_j^A)^N V^{(N)}(A_j), \quad V^{(N)}(B_{\text{tot}}) = \sum_{k=1}^{r_B} (\mu_k^B)^N V^{(N)}(B_k), \quad (1)$$

written $V^{(N)}(\text{toTensorFromBlocks}\mu A)$ in the formalization.¹ A cross-overlap $\langle V^{(N)}(A_j), V^{(N)}(B_k) \rangle$ is `MPSTensor.mpvOverlap`. *Eventual proportionality* of the assembled families is the hypothesis that for some scalar sequence $c_N \neq 0$ cofinally,

$$\sum_j (\mu_j^A)^N V^{(N)}(A_j) = c_N \sum_k (\mu_k^B)^N V^{(N)}(B_k) \quad (\forall^{\text{cof}} N), \quad (2)$$

formalized as `EventuallyNonzeroProportionalMPV2`.

The surviving one-copy normal-canonical BNT formulation `IsNormalCanonicalFormBNT` carries a single weight μ_k per sector with strict ordering

$$\|\mu_1\| > \|\mu_2\| > \cdots > \|\mu_r\| \quad (\text{mu_strict_anti})$$

across the entire family.² Together with the dominant normalization $\|\mu_1\| = 1$, every sub-dominant block has modulus strictly less than 1. [CPGSV21, Definition 4.2 and lines 1864–1884] states the corresponding source structure as a two-layer canonical form: spectral levels λ_j with $\|\lambda_1\| > \cdots > \|\lambda_g\|$ factored from within-sector unit-modulus weights $\mu_{j,q}$ ($\|\mu_{j,q}\| = 1$) with sector multiplicities $r_j \geq 1$. The `IsNormalCanonicalFormBNT` formulation is the one-copy specialization $r_j = 1$ of that structure; see `docs/paper-gaps/ft_one_copy_scope_restriction.tex`.

¹`MPSTensor.toTensorFromBlocks`; `MPSTensor.mpv.toTensorFromBlocks_eq_sum`.

²`MPSTensor.IsNormalCanonicalFormBNT.mu_strict_anti` in `TNLean/MPS/BNT/Construction`.

2 Cited assertion

The block-selection paragraph in the proof of Theorem `thm1` [CPGSV16, line 1182] fixes any B -block B_{k_0} and rules out

$$\langle V^{(N)}(A_j), V^{(N)}(B_{k_0}) \rangle \xrightarrow{N \rightarrow \infty} 0 \quad (\text{all } j), \quad (\mathcal{H}_{k_0})$$

by Corollary `Lem1` [CPGSV16, line 1131], which gives eventual linear independence of $\{V^{(N)}(A_j)\}_j \cup \{V^{(N)}(B_{k_0})\}$ from $(\mathcal{H}_{k_0})$ and combines it with the proportionality identity at [CPGSV16, eq. `eq:II_ABasicTensors`, line 286].

3 Formal target

The mathematical target is, for $k_0 : \text{Fin } r_B$ arbitrary: given the one-copy normal-canonical BNT hypotheses on both μ^A, A and μ^B, B , `EventuallyNonzeroProportionalMPV2` between the assembled tensors, an index $k_0 : \text{Fin } r_B$ and the hypothesis $\forall j, \langle V^{(N)}(A_j), V^{(N)}(B_{k_0}) \rangle \rightarrow 0$, derive $\perp$. The old one-copy fixed-block declarations for this target have been retired. The surviving route is the `SectorBNT` proportional matching layer, which keeps the raw coefficients of `SectorDecomposition` visible and uses the API lemmas in `TNLean/MPS/FundamentalTheorem/SectorBNT/Api`. The universally quantified fixed-block cancellation remains a mathematical obligation.

4 Where the per-block projection fails for non-dominant k_0

Projecting (2) onto $V^{(N)}(B_{k_0})$ gives, with $\text{LHS}_N := \langle V^{(N)}(A_{\text{tot}}), V^{(N)}(B_{k_0}) \rangle$ and $\text{RHS}_N := c_N \langle V^{(N)}(B_{\text{tot}}), V^{(N)}(B_{k_0}) \rangle$,

$$\text{LHS}_N = \sum_j (\mu_j^A)^N \langle V^{(N)}(A_j), V^{(N)}(B_{k_0}) \rangle, \quad \text{RHS}_N = c_N \sum_k (\mu_k^B)^N \langle V^{(N)}(B_k), V^{(N)}(B_{k_0}) \rangle. \quad (3)$$

With $\|\mu_j^A\| \leq 1$, every $(\mu_j^A)^N$ is bounded; together with $(\mathcal{H}_{k_0})$ this forces $\text{LHS}_N \rightarrow 0$. For RHS_N , the current cross-overlap decay statement `MPSTensor.IsBNTCanonicalForm.cross_overlap_basis_tendsto_zero` gives $\langle V^{(N)}(B_k), V^{(N)}(B_{k_0}) \rangle \rightarrow 0$ for $k \neq k_0$, while the diagonal block has a nonzero limiting coefficient, so $\text{RHS}_N = c_N (\mu_{k_0}^B)^N (\ell + o(1))$. In the dominant block, the proportionality identity and the BNT normalizations give the dominant-adjusted scalar limit $\|c_N (\mu_0^B / \mu_0^A)^N\| \rightarrow 1$. Therefore

$$\|c_N (\mu_{k_0}^B)^N\| \sim \|\mu_0^A\|^N \|\mu_{k_0}^B / \mu_0^B\|^N. \quad (4)$$

Under $\|\mu_0^A\| = 1$ and `mu_strict_anti`, the second factor in (4) decays geometrically whenever $k_0 \neq 0$, so $\text{RHS}_N \rightarrow 0$ together with LHS_N . No contradiction is

available from the per-block projection. The dominant case $k_0 = 0$ is the unique index for which the ratio in (4) is the constant 1 and the bound stays uniform; that is the analytic content that the dominant fixed-block statement captured before the retirement of the old one-copy layer.

Two independent probes confirm the obstruction. Path α (`audits/2026-05-13_cpsv16_ft_exact_leading_coeff.md`, PR #1654) shows an exact eventual leading-coefficient identity $c_N(\mu_0^B)^N \zeta^N = (\mu_0^A)^N$ fails at $r_A = r_B = 2$ with distinct sub-dominant moduli. Path β scope adjustment (`audits/2026-05-13_cpsv16_ft_path_beta_scout.md`, PR #1664) shows the lifted per-block projection on the `SectorDecomposition` surface inherits the same decay mismatch.

5 The argument the source uses

The Step 1 contradiction in the proof of Theorem `thm1` [CPGSV16, line 1182] does not project onto $V^{(N)}(B_{k_0})$. Isolating $V^{(N)}(B_{k_0})$ in (2) gives

$$\sum_j (\mu_j^A)^N V^{(N)}(A_j) - c_N \sum_{k \neq k_0} (\mu_k^B)^N V^{(N)}(B_k) = c_N (\mu_{k_0}^B)^N V^{(N)}(B_{k_0}), \quad (5)$$

whose left-hand side lies in $\text{span}(\{V^{(N)}(A_j)\}_j \cup \{V^{(N)}(B_k) : k \neq k_0\})$. A larger application of `Lem1` provides eventual linear independence of the combined family with full span dimension,

$$\dim \text{span}(\{V^{(N)}(A_j)\}_{j=1}^{r_A} \cup \{V^{(N)}(B_k)\}_{k=1}^{r_B}) = r_A + r_B \quad (\forall^{\text{cof}} N), \quad (6)$$

under all-pairwise cross-overlap decay. With (6), (5) forces $c_N(\mu_{k_0}^B)^N = 0$ by coefficient identification, contradicting $c_N \neq 0$ and $\mu_{k_0}^B \neq 0$. The mechanism uses only the coefficient match forced by linear independence; it requires neither side to stay bounded away from 0.

The current `SectorBNT` API provides the combined-family linear-independence mechanism as `MPSTensor.lsBNTCanonicalForm.combined_family_eventually_li`. The remaining task is to combine that statement with the proportional coefficient identity in the fixed-block setting, without appealing to the retired per-block-projection layer.

6 Classification

Scope restriction combined with an open gap. The retired dominant fixed-block declarations proved the Step 1 conclusion for $k_0 = \langle 0, \cdot \rangle$ on the one-copy surface. The universally quantified statements stay open because the cited mechanism requires either

1. the combined-family `Lem1` statement (6) with all-pairwise cross-overlap decay, enabling the coefficient-identification step of (5); or

2. the two-layer CPSV structure of [CPGSV21, Definition 4.2 and lines 1864–1884], separating spectral levels λ_j from within-sector unit-modulus weights $\mu_{j,q}$. On the surviving `SectorDecomposition` surface this means keeping the raw coefficients $\sum_q \mu_{j,q}^N$ visible and using `MPSTensor.lsBNTCanonicalForm.coeff_not_tendsto_zero_at_block` rather than collapsing the sector to a single weight before the projection argument. Tracked in GitHub issue #1641.

The collapse $r_j = 1$ enforced by `mu_strict_anti` identifies the two layers into a single weight; once collapsed, the non-dominant Step 1 conclusion cannot be reached by per-block projection under the one-copy normal-canonical BNT formulation because both sides of the projected proportionality decay together.

References

- [CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: *Annals of Physics* 378 (2017), 100–149, 2016.
- [CPGSV21] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product states and projected entangled pair states: Concepts, symmetries, theorems. *Reviews of Modern Physics*, 93(4):045003, 2021.